

12 The number of insects in a population at the start of the year n is P_n

The number of insects in the population at the start of year $(n + 1)$ is P_{n+1} where

$$P_{n+1} = kP_n$$

Given that k has a constant value of 1.13

- (a) find out how many years it takes for the number of insects in the population to double.
You must show how you get your answer.

(2)

The value of k actually increases year on year from its value of 1.13 in year 1

- (b) How does this affect your answer to part (a)?

(1)

(Total for Question 12 is 3 marks)